

Refining the Computational Analysis of Interactive Fiction: A Hybrid Framework using Bag-of-Paths Formalism and LLMs

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Introduction

Interactive Fictions and Gamebooks

An **interactive fiction** is a form of narrative in which the reader's or player's choices influence the sequence of events, the accessible content, or the story's outcome.

A **gamebook** is a book-based form of interactive fiction, usually divided into numbered sections connected by choices, rules, and sometimes game mechanics such as combat, inventory, or chance.

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Suddenly, the large rock you are hiding behind is rolled aside and you are faced by two snarling Giaks intent on your death. The cave mouth is a narrow entrance and you can only fight the Giaks one at a time.



Giak 1: COMBAT SKILL 13 ENDURANCE 10

Giak 2: COMBAT SKILL 12 ENDURANCE 10

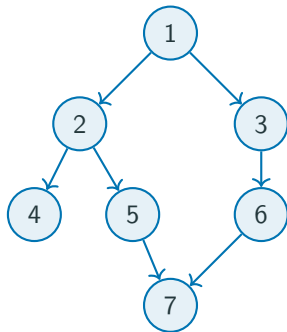
If you win, you may explore the cave further by **turning to 33**.

Or you may leave and descend the hill. **Turn to 248**.

The classical representation: a graph

Interactive fiction is traditionally modeled as a **directed graph**.

- **Nodes** represent *paragraphs*, i.e. numbered narrative sections.
- **Edges** represent *transitions*, i.e. the choices available to the player.



The **Main limitation** of this representation :

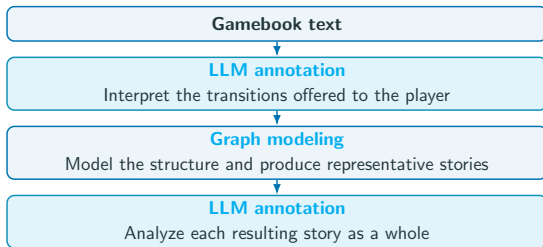
- The **structure** or the **topology** of the work can be studied.
- The text itself, therefore the **narrative meaning**, is no longer taken into account.

A hybrid solution: graphs and LLMs

Large Language Models (LLM) make it possible to reintroduce the semantic content discarded by graph modeling.

Our approach : both methods

- The **graph** models the **structure** of the work
- The **LLM** interprets the **narrative content**



Research question

How can graph modeling and local LLM annotation be combined to study both the structural possibilities and the narrative experiences produced by a gamebook?

Methodological choice : local LLM

Methodological choice

- We are going to use exclusively [Qwen3.6-27B](#), executed locally on university infrastructure.
- The local model is not treated as an autonomous literary critic. It is used as a [controlled annotation instrument](#) whose decisions remain open to inspection.

Methodological reasons

- No dependency on a proprietary API
- Prompts and inference settings are documented
- Outputs can be stored, inspected and reproduced
- Source texts remain on local infrastructure
- Lower and predictable financial cost

Why is it sufficient?

- The task is a constrained annotation task
- Categories are defined by an explicit codebook
- Outputs follow a restricted JSON schema
- Results are checked against human annotations
- Frontier-level generative capabilities are overkill

Corpus: Lone Wolf and Project Aon

The studied corpus is the **Lone Wolf series** created by Joe Dever.

The **project Aon** (www.projectaon.org) provides authorized digital editions of the books. It is already structured with html.

There are **several mechanics** in these gamebooks :

- Branching choices and forced transitions
- Random events
- Player profile with stats and special powers
- Combat with possible evasion
- Endurance (HP) management
- Items and inventory management
- Puzzles and hidden links

Scope of this presentation

A complete representation of every mechanic would make the model intractable. We use **explicit simplifications** and focus exclusively on **Book 1: Flight from the Dark**.

Modeling scope: from L0 to L3

The goal is to build a tractable **Markovian model**: for a fixed player profile, the next transition depends only on the current node, not on the complete play history.

Level	Information	Mechanics	Decision
L0	Topology	Paragraphs, links, forced transitions, Death and Win outcomes	Included
L1	Player agency	Choices annotated along the risk, morality and action axes	Included
L2	Uncertainty	Random events, combat, evasion, Kai Disciplines (special powers) and conditions	Simplified
L3	Persistent state	Endurance, healing, inventory, gold, equipment and meals	Not simulated

L2 simplifications :

- One global combat-win probability
- One global escape probability
- Kai Disciplines availability fixed at 0.5
- One generic `has_condition` probability

Outline of the presentation

1. Paragraph parsing and transition annotation

A local LLM reads the choices offered at each paragraph and outputs :

- Transition types
- Game mechanics
- Risk, morality and action labels

2. Graph modeling and Bag-of-Path formalism

The annotations are compiled into weighted graphs for different player profiles, and various objects are computed :

- Bag-of-Paths indices
- Local and global measures
- Representative trajectories

3. Trajectory annotation

A local LLM reads the stories associated with selected trajectories and produce :

- Whole-story annotations
- Profile manifestation
- Pairwise comparisons

Paragraph parsing and transition annotation

Objective: from paragraphs to structured data

Objective

Transform each gamebook paragraph into an **structured representation**, separating **narrative units** from the **transitions** offered to the player.

Project Aon page

112

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Paragraph 112



Parsing

+

LLM

LW01_nodes.csv

node_id	text_content	absorbing_status	enemies
112	Suddenly, the large rock...	potential_death	Giak 1; Giak 2

LW01_edges.csv

source_id	target_id	edge_text	transition_type	realisation_value
112	33	If you win, explore the cave...	explicit_choice	If you win
112	248	Leave and descend the hill...	explicit_choice	If you win

source_id	target_id	semantic_risk	semantic_morality	semantic_action	warnings
112	33	reckless	neutral	tactical	-
112	248	cautious	selfish	physical	-

Rule-based parsing versus LLM-based annotations

Rule-based parsing (on html)

Can process all fields in the **node table**, and **some fields in the edges tables** (source_id, target_id, and edge_text, used for validation).

LLM-based annotations

All fields in the **edge table**, in particular :

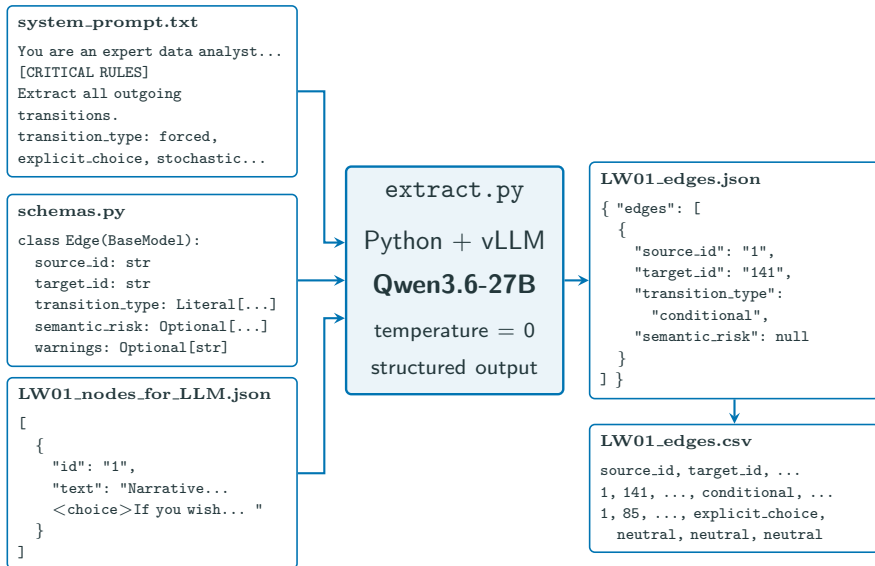
- Transition mechanism: transition_type, realisation_value;
- Meaning of player choices: semantic_risk, semantic_morality, semantic_action;
- Ambiguities requiring attention: warnings.

Methodological choice

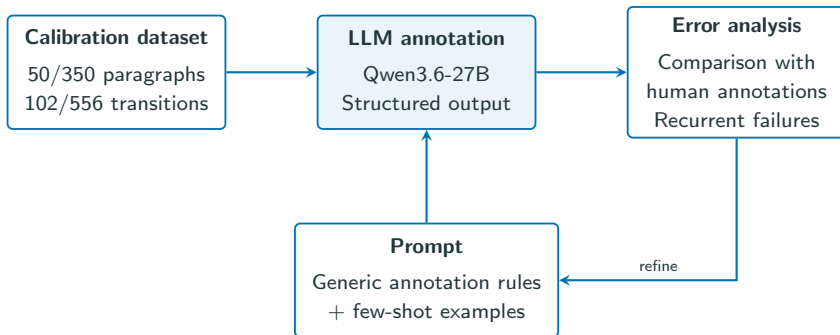
Each explicit player choice is annotated according to **3 independent semantic axes** :

- **Risk**: exposure to danger (cautious, neutral, reckless)
- **Morality** : treatment of others (selfish, neutral, noble)
- **Action** : mode of intervention (physical, neutral, tactical)

From prompt to structured annotation



Iterative prompt refinement



- Errors lead to revisions of **general annotation rules**, not paragraph-specific corrections.
- This iterative process is a **calibration**, not an independent evaluation.
- No held-out validation set was used because of limited time; generalization to another book is therefore not measured.

Annotation results

Calibration : On 50 paragraphs and 102 transitions, the final prompt produced no missing or invented transition and no transition-type disagreement. Only 3 (on 177 annotations) semantic disagreements : two for risk and one for action.

Paragraph statistics

Total	350
Terminal	16 deaths / 1 victory
One outgoing transitions	157
Two outgoing transitions	134
Three or more outgoing transitions	42
Paragraphs with enemies	29

Transition statistics

Transition type	N	%
Explicit choice	291	52.3
Forced	157	28.2
Conditional	66	11.9
Stochastic	42	7.6
Total	556	100.0

Axis	First pole	Neutral	Opposite pole
Risk	cautious: 95 (32.6%)	84 (28.9%)	reckless: 112 (38.5%)
Morality	selfish: 18 (6.2%)	262 (90.0%)	noble: 11 (3.8%)
Action	physical: 101 (34.7%)	126 (43.3%)	tactical: 64 (22.0%)

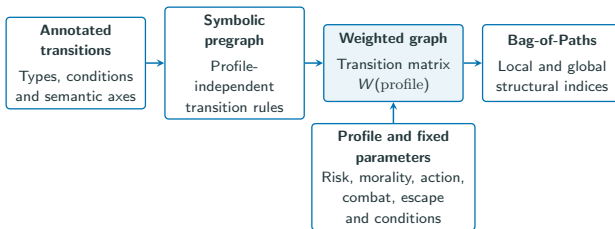
Warnings	53 (9.5%)
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Graph construction and Bag-of-Paths analysis

Objective: from annotations to structural measures

The objectives here are to :

- Transform annotations into **weighted directed graphs (i.e. a Markov chain matrix)** which depend on **player profiles**.
- Compute **various indices with the Bag-of-Path formalism**.



This section addresses three questions:

- How can heterogeneous game mechanics be converted into transition probabilities?
- How does the resulting graph vary across player profiles?
- Which paragraphs and structural properties are most important across all possible trajectories?